

Math 140 Spring 2023 - Worksheet 4.2

Section:

Name:

1. Find all numbers c in the interval $(-2, 1)$ for which the line tangent to the graph of $f(x) = x^3 + 4$ is parallel to the line joining $(-2, f(-2))$ and $(1, f(1))$.
2. Let $f(x) = |x|$. Notice that $f(-2) = f(2)$, but there is no number c in $(-2, 2)$ such that $f'(c) = 0$. Does this result contradict Rolle's Theorem? Why?
3. Suppose f is differentiable at every number in $[-1, 1]$, and $f'(1) = f'(-1) = 1$. Assume in addition that $f(-1) = f(1) = 0$. Explain why f has at least one zero in $(-1, 1)$. Using this result, f' should have at least two zeros in $(-1, 1)$. Why?